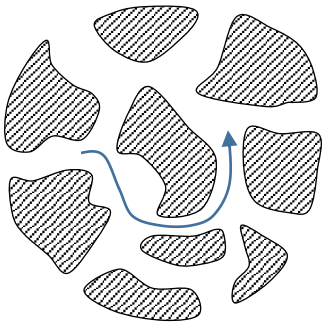
CEBT II REACTION ENGINEERING

6. Reactions in porous solids

- For many heterogeneous systems, reactions take place within a porous catalyst.
- Up until now, we have assumed that one step (be it intrinsic chemical kinetics, internal mass transport, or external mass transport) is the rate limiting one.
- We now consider the case that both internal diffusion and chemical kinetics affect the rate of reaction.
  - In a few cases, analytical solutions exist for the overall rate of reaction (e.g. when the chemical kinetics are first order). These will give us insight into the behaviour for those cases where there is no analytical solution.
  - In other cases, numerical solutions will need to be obtained.

6.1 Diffusion within porous solids

- Molecular transport in porous solids is an active area of research (particularly for mixtures and if two phases are present).



- We shall use an effective diffusion coefficient to describe transport in unit volume of the particle (which contains both pore space and a solid fraction)

$$D_e = \frac{\varepsilon_{particle} D_{pore}}{\tau}$$

where

D_{pore} is the diffusion constant within the pore space alone

$\varepsilon_{particle}$ is the porosity of the particle (typically 0.2-0.7 for a catalyst)

$$\varepsilon_{particle} = \frac{\text{pore volume of particle}}{\text{total volume of particle}}$$

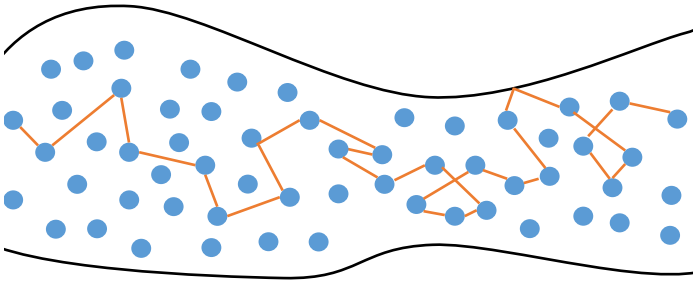
τ is the tortuosity of the pore network (typically 2-6 for a catalyst)

$$\tau = \frac{\text{distance a molecule needs to travel between two points}}{\text{straight line distance between two points}}$$

- This equation applies for both liquids and gases in porous solids.
- For gases, we can consider further the diffusion constant D_{pore} within the pore space; there are three limiting cases.
    1. Molecular diffusion
    2. Knudsen diffusion
    3. Surface diffusion
- We shall consider the characteristics of each limiting case in turn.

6.1.1 Molecular diffusion

- In this case, diffusion is controlled by molecular collisions between gas-phase molecules. This occurs
    - when the mean free path between gas-phase collisions is less than the pore size; this will be the case at high pressures and/or with large pores.
    - in the absence of strong adsorption sites.
- With molecular diffusion, the pore space diffusion coefficient will be the same as for the bulk gas phase.



Molecular diffusion within a pore

- For a single species, the kinetic theory of gases predicts

$$D_{\text{bulk}} = \frac{1}{3}\lambda\bar{c} = \frac{1}{3}\frac{k_{\text{B}}T}{\sqrt{2}\sigma P}\sqrt{\frac{8k_{\text{B}}T}{\pi m}}$$

  λ is the mean free path

  $\bar{c}$ is the mean speed

  σ is the cross-sectional area of the molecule

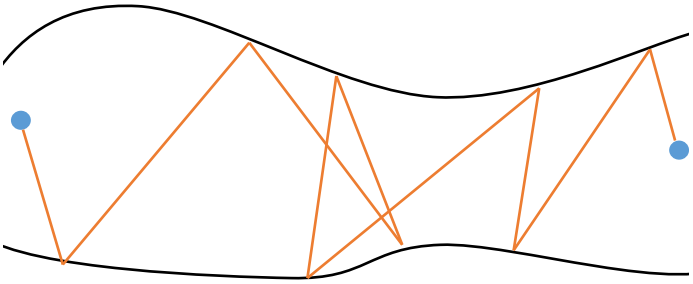
  k_{B} is Boltzmann's constant

  m is the mass of a molecule

    - For greater accuracy semi-empirical correlations may be used, and adjustments made for mixtures.

6.1.2 Knudsen diffusion

- In this case, diffusion is controlled by collisions between gas-phase molecules and the pore walls. This occurs:
  - when the mean free path between gas-phase collisions is larger than the pore size; this will be the case at low pressures and/or with small pores.
  - in the absence of strong adsorption sites.



Knudsen diffusion within a pore

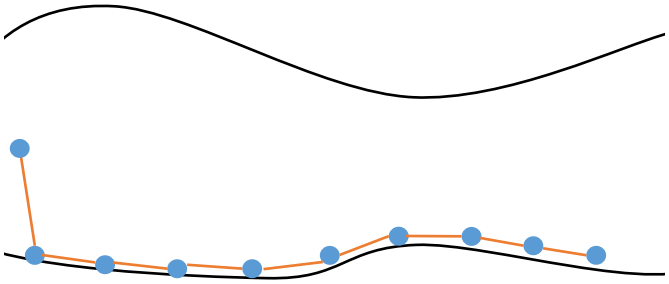
- We replace the mean free path in the earlier expression by the pore diameter (to reflect the distance between collisions).

$$D_{\text{Knudsen}} = \frac{1}{3} d_{\text{pore}} \bar{c} = \frac{1}{3} d_{\text{pore}} \sqrt{\frac{8 k_{\text{B}} T}{\pi m}}$$

where d_{pore} is the pore diameter.

6.1.3 Surface diffusion

- For surface diffusion, transport is along the pore wall – this may occur if there are strong adsorption sites.



Surface diffusion within a pore

- Molecules need to overcome an activation energy before they can "hop" to the neighbouring site.

$$D_{\text{surface}} = D_0 \exp\left(\frac{-E_{\text{act,diff}}}{RT}\right)$$

- D_{surface} usually has a strong temperature dependence.

- Surface diffusion may also occur if the pores are so small that they are of molecular dimensions (even in the absence of strong adsorption sites).
  - In that case adsorbed molecules will always be in contact with the surface.
  - For medium-sized molecules, this case is sometimes called configurational diffusion.

6.1.4 Overall pore diffusion coefficient

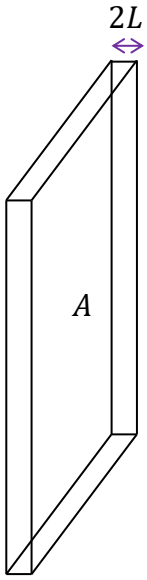
- The overall pore space diffusion coefficient may be estimated by

  $D_{\text{pore}} = D_{\text{bulk}} + D_{\text{Knudsen}} + D_{\text{surface}}$

  - Here we are dealing with diffusion resistances in parallel, so the smallest resistance dominates. (For diffusion resistances in series, the resistances (i.e. the reciprocals of diffusion coefficients) add and so the largest resistance dominates.)
- Different experimental techniques for measuring diffusion tend to give different results. This is because different techniques look at transport for different timescales and thus different lengthscales.
  - Lengthscale of a single pore
  - Lengthscale of a single particle (in which case many pores are involved),
  - Lengthscale of many particles (in which case what happens at the external surface of the particles becomes important).

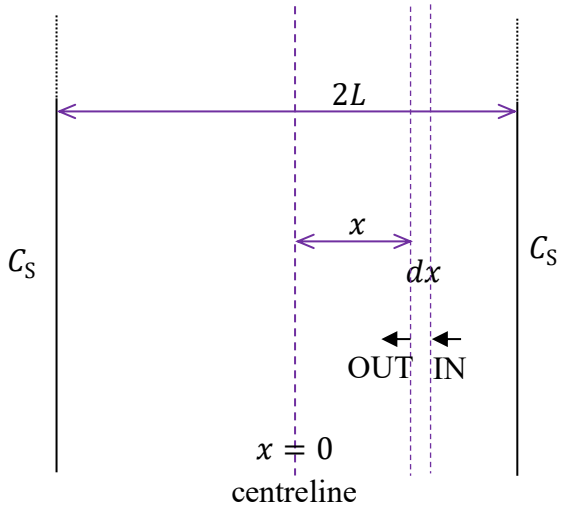
6.2 Analysis of chemical reaction with internal diffusion

- For convenience, we shall consider the simplest case to model mathematically first and then consider extensions.



- The catalyst is a thin porous slab of thickness $2L$ and large area A.
- The reagent transports within the catalyst with an effective diffusion coefficient D_e.
- The reagent reacts by first-order kinetics with rate constant k_V (based on unit volume of catalyst).
- Isothermal behaviour is assumed.

6.2.1 Concentration profile for a first-order reaction within a porous slab



- The reagent has concentration C_S at both faces of the slab.
- There will be a concentration profile $C(x)$ within the catalyst pellet.
- Consider a mass balance on the volume element between x and $x + dx$ from centreline
- IN – OUT term:

$$AD_e \frac{d(C + dC)}{dx} - AD_e \frac{dC}{dx} = AD_e \frac{d^2C}{dx^2} dx$$

  - The sign used here is for the direction of flow indicated
  - We have assumed D_e is independent of x (which is expected to be the case unless the pore space is different in different parts of pellet).
- MADE term:

$$r_V \, A \, dx = -k_V \, C \, A \, dx$$

- ACCUMULATED term:

$$\frac{dC}{dt} A \, dx$$

- At steady state, the mass balance is therefore

$$AD_e \frac{d^2C}{dx^2} dx - k_V C A dx = 0$$

$$\frac{d^2C}{dx^2} - \frac{k_V}{D_e}C = 0$$

- This is a straightforward 2nd order ODE to solve (provided the diffusion coefficient does not depend on concentration). The solution is

$$C = a\exp\left(\sqrt{\frac{k_V}{D_e}}x\right) + b\exp\left(-\sqrt{\frac{k_V}{D_e}}x\right)$$

- The boundary conditions are

$$C = C_S \quad \text{at} \quad x = \pm L$$

$$\frac{dC}{dx} = 0 \quad \text{at} \quad x = 0$$

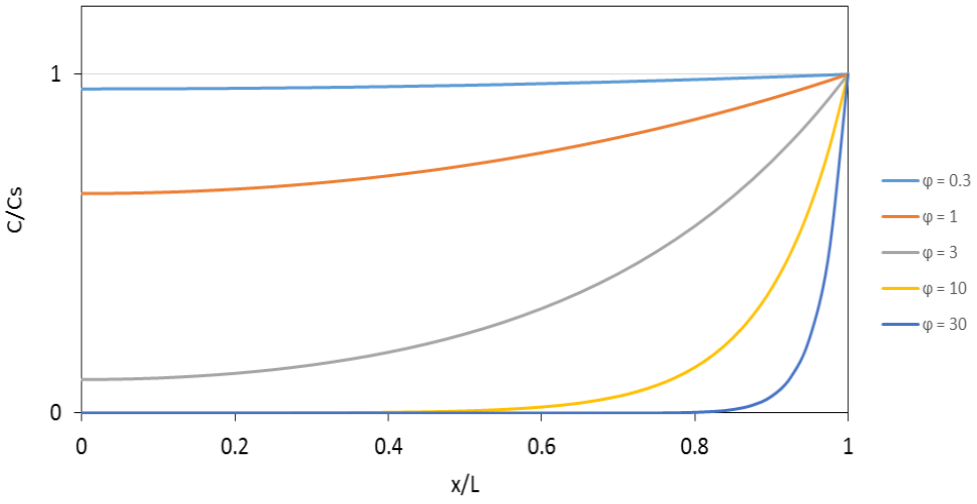
- This leads to the solution is

$$C = C_S\frac{\cosh(\varphi\, x/L)}{\cosh\varphi}$$

where

$$\varphi = L\sqrt{\frac{k_V}{D_e}}$$

- The parameter φ is termed the "Thiele modulus".
  - The Thiele modulus is a dimensionless number quantifying the relative magnitudes of chemical reaction rate and internal diffusion rate.
- In this case, profiles of concentration within the catalyst pellet at different values of Thiele modulus are shown in the graph below.



Concentration profiles for a first-order reaction within a slab of catalyst

- At low Thiele modulus:
  - The reaction rate is relatively slow, internal diffusion is relatively fast.

- The reagent can penetrate to the centre of the catalyst pellet.
- In the limit of very low φ, the concentration profile within the catalyst will be close to uniform (i.e. $C \approx C_S$ for all x).

- At high Thiele modulus:
  - The reaction rate is relatively fast, internal diffusion is relatively slow.
  - The reagent cannot penetrate to the centre of the catalyst pellet – it will react mainly in the outer regions of the pellet.

6.2.2 Effectiveness factor for a first-order reaction within a porous slab

- Define an internal effectiveness factor η. Equivalent definitions:

$$\eta = \frac{\text{average rate of reaction with (heat and) mass transfer limitations}}{\text{rate of reaction evaluated at surface conditions}}$$

$$\eta = \frac{\text{mol/s reacting in catalyst}}{\text{mol/s that would react if whole catalyst was at surface conditions } (T_S, C_S)}$$

$$\eta = \frac{r_{\text{overall}}}{r_{\text{no internal transport limitations}}}$$

- The internal effectiveness factor is useful because the reaction engineer can use the familiar design equations (e.g. Section 1), but with an overall rate of reaction given, for the first-order case, by

$$r_{\text{overall}} = \eta\, k_V C_S$$

- We can derive an analytical expression for η for the special case considered (slab, 1st order).
- The overall mol/s reacting in the catalyst is

$$r_{\text{overall}} = \int_{-L}^{L} k_V\, C\, A\, dx$$

$$r_{\text{overall}} = 2\int_{0}^{L} k_V\, C_S \frac{\cosh(\varphi\, x/L)}{\cosh\varphi}\, A\, dx$$

(using symmetry to simplify the maths slightly)

$$r_{\text{overall}} = \frac{2k_V C_S A}{\cosh\varphi}\left[\frac{L}{\varphi}\sinh(\varphi\, x/L)\right]_0^L$$

$$r_{\text{overall}} = 2k_V C_S AL\frac{\tanh\varphi}{\varphi}$$

  - Aside: this could also be derived from the rate that molecules of reagent enter the catalyst (as that will be equal to the overall rate of reaction at steady state).

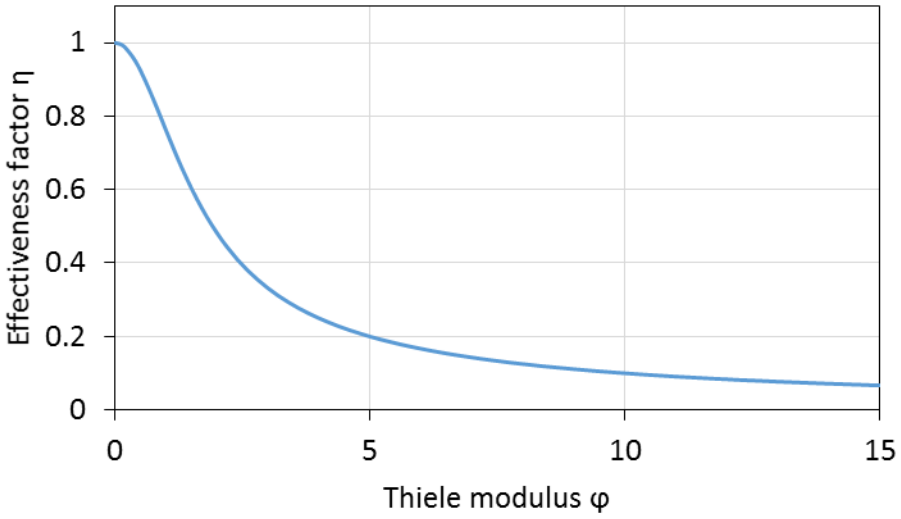
- The mol/s that would react if all the catalyst was at surface conditions is

$$r_{\text{no internal transport limitations}} = 2k_V C_S AL$$

- The internal effectiveness factor in this case is therefore

$$\eta = \frac{r_{\text{overall}}}{r_{\text{no internal transport limitations}}}$$

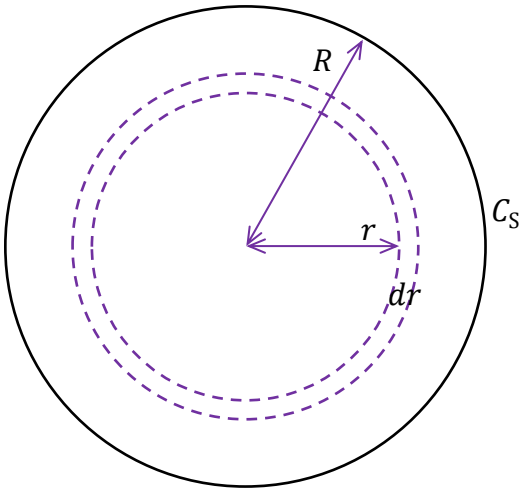
$$\eta = \frac{\tanh \varphi}{\varphi}$$



Internal effectiveness factor for a first-order reaction within a slab of catalyst

6.2.3 Concentration profile for a first-order reaction within a porous sphere

- Other catalyst geometries can be considered (e.g. spheres, cylinders) rather than just slabs.
  - The derivations are similar, but the maths is a bit more complex.
- For the case of a sphere



$$-\frac{d}{dr}\left(-4\pi r^2 D_e \frac{dC}{dr}\right) dr - k_V C 4\pi r^2 dr = 0$$

$$D_e\left(\frac{d^2C}{dr^2} + \frac{2}{r}\frac{dC}{dr}\right) - k_V C = 0$$

- The boundary conditions are

  $C = C_S$ at $r = R$

  $\frac{dC}{dr} = 0$ at $r = 0$

- The solution (not proven here, but derivation is possible using the substitution $C = y/r$) is

  $$C = C_S \frac{\sinh(\varphi r/R)}{(r/R)\sinh\varphi}$$

  with

  $$\varphi = R\sqrt{\frac{k_V}{D_e}}$$

- The solution leads to an internal effectiveness factor for a sphere with 1st order reaction of

  $$\eta = \frac{3}{\varphi}\left(\frac{1}{\tanh\varphi} - \frac{1}{\varphi}\right)$$

  - η tends to 1 as φ tends to zero
  - With this form of Thiele modulus, η tends to $3/\varphi$ as φ becomes large.

6.2.4 Generalised Thiele modulus

- There is a choice of length parameter used in the definition of Thiele modulus.
  - For mathematical convenience, in derivation above we used half-thickness of a slab.
  - For mathematical convenience, in derivation above we used R for a sphere.
- Consider defining a generalised Thiele modulus for a first-order reaction as

  $$\varphi_{1,gen} = \frac{V_{cat}}{A_{ext}}\sqrt{\frac{k_V}{D_e}}$$

  - The length parameter is now Volume / External Surface Area of the pellet.
  - For a slab: the length parameter is L (when the slab thickness is $2L$).
  - For a sphere: the length parameter is $R/3$ (when the radius is R).
- It turns out that, with this definition, the effectiveness factor obeys

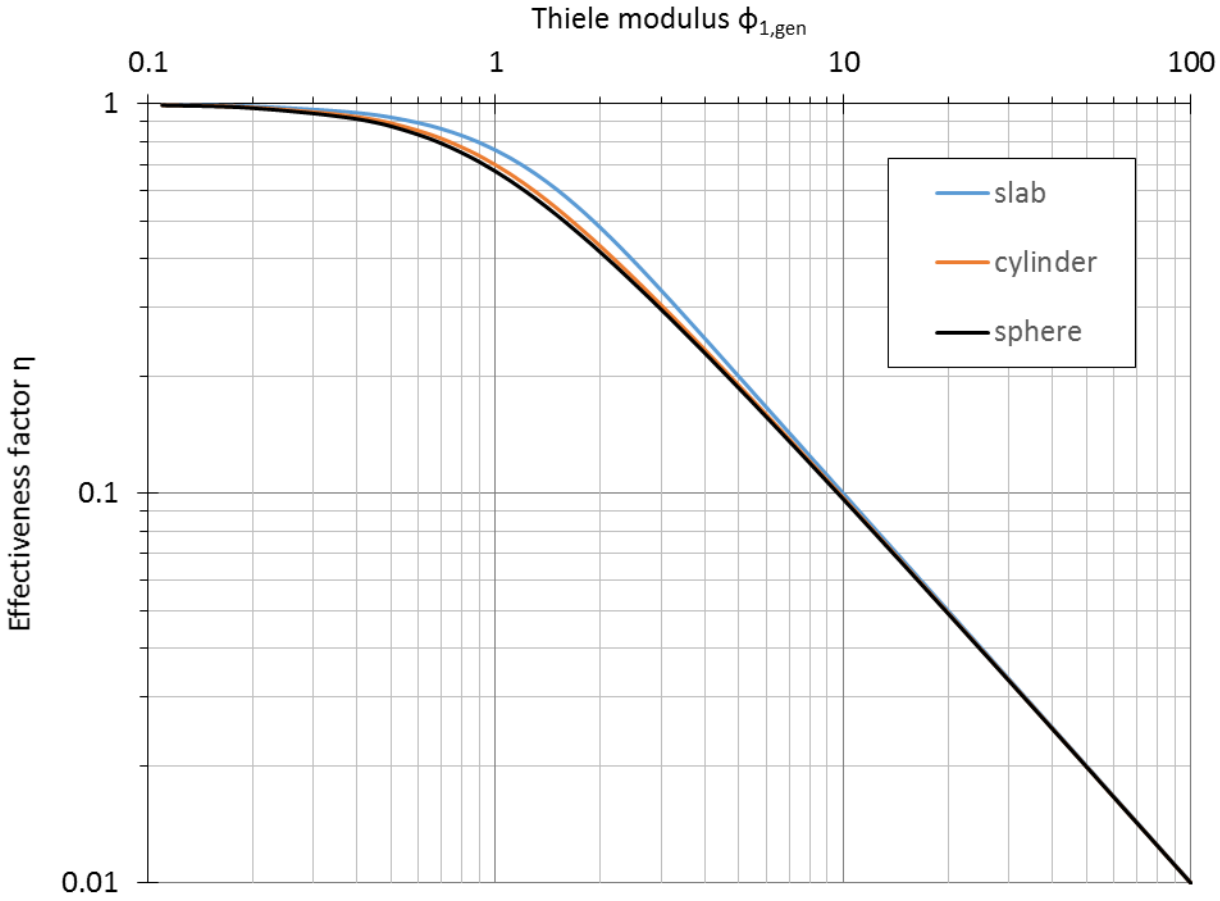
  $$\eta \approx \frac{1}{\varphi_{1,gen}} \quad \text{for large } \varphi_{1,gen}$$

  - This is true regardless of whether the pellet is a slab, cylinder, sphere or other shape.

- For instance, the effectiveness factor for a first-order reaction in a sphere is (using generalised Thiele modulus):

$$\eta = \frac{1}{\varphi_{1,\text{gen}}}\left(\frac{1}{\tanh\left(3\varphi_{1,\text{gen}}\right)} - \frac{1}{3\varphi_{1,\text{gen}}}\right)$$

- The analytical results for first-order reactions in a slab, a long cylinder, and a sphere are shown below.



Internal effectiveness factor for a first-order reaction within different shape catalyst pellets

- Note:
  - this graph uses logarithmic axes
  - when using the generalised Thiele modulus, we get similar behaviour for all three catalyst pellet shapes considered.
  - we can use this graph to predict effectiveness factors for any shape pellet; most real pellets will show behaviour between that of a cylinder and sphere.

6.2.5 Other rate laws

- It is only possible to obtain an analytical solution for concentration profile within a catalyst for the 1st order case (as above) and the 0th order case (as on the Problem Sheet).
- For an nth order reaction, the generalised dimensionless Thiele modulus is

$$\varphi_{n,\text{gen}} = \frac{V_{\text{cat}}}{A_{\text{ext}}}\sqrt{\frac{(n+1)k_{\text{V}}C_{\text{S}}^{n-1}}{2D_{\text{e}}}}$$

- With this definition, qualitatively similar behaviour to first-order reactions is observed.
  - At low values of Thiele modulus, η tends to 1
  - At high values of Thiele modulus, η tends to $1/\varphi_{n,\text{gen}}$
- Consider a second-order reaction ($n = 2$):
  - If C_{S} is increased, then $\varphi_{n,\text{gen}}$ increases, and so η decreases.
  - Hence any diffusional limitations become more significant if C_{S} is increased.

6.2.6 Non-isothermal behaviour

- So far, we have assumed isothermal behaviour – the temperature within the pellet is constant.
- In reality, the reaction will be exothermic or endothermic, and so there may be a temperature profile that depends on where the reaction takes place and the rate of heat transport. See Section 6.3.
- For an exothermic reaction:
  - The temperature may be higher within the catalyst than at its surface.
  - The rate constant k will be higher where the temperature is higher.
  - We will get higher internal effectiveness factors than expected; indeed, they may even exceed unity as they are defined based on T_{S}.

6.2.7 Disguised kinetics

- The term "disguised kinetics" is used when the overall kinetic behaviour does not show the same behaviour as the intrinsic kinetics because of mass or heat transfer limitations.
  - The measured reaction order may not reflect the true order of the chemical reaction.
  - The measured activation energy may not reflect the true activation energy of the chemical reaction.
- Consider an nth order reaction with diffusional limitations. In this case

$$r_{\text{overall}} = \eta\, k_{\text{V}} C_{\text{S}}^{n}\, V_{\text{cat}}$$

$$\varphi_{n,\text{gen}} = \frac{V_{\text{cat}}}{A_{\text{ext}}}\sqrt{\frac{(n+1)k_{\text{V}}C_{\text{S}}^{n-1}}{2D_{\text{e}}}}$$

- If diffusion is the limiting factor, the Thiele modulus is large and we know $\eta \approx 1/\varphi_{n,\text{gen}}$. In that special case

$$r_{\text{overall}} = \frac{A_{\text{ext}}}{V_{\text{cat}}}\sqrt{\frac{2D_{\text{e}}}{(n+1)k_{\text{V}}C_{\text{S}}^{n-1}}}\,(k_{\text{V}}C_{\text{S}}^{n})\,V_{\text{cat}}$$

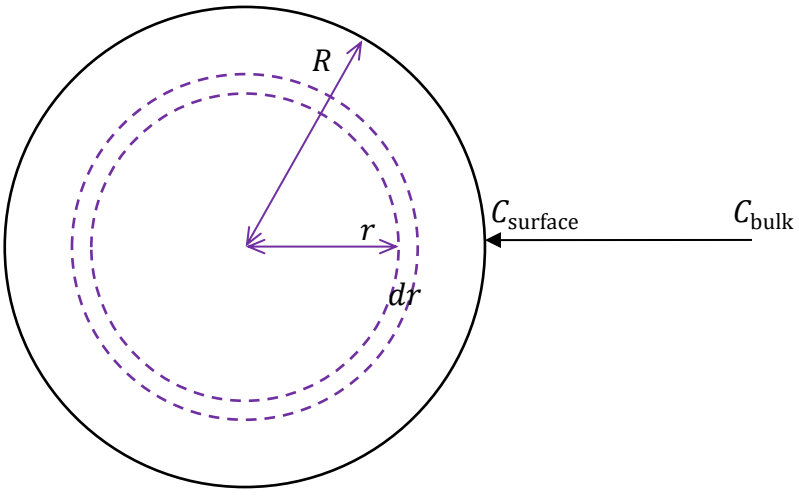
$$r_{\text{overall}} = A_{\text{ext}}\sqrt{\frac{2D_{\text{e}}k_{\text{V}}}{(n+1)}}\;C_{\text{S}}^{(n+1)/2}$$

- So the overall reaction will have an apparent order of $(n+1)/2$ when diffusion is a limiting factor.
- Temperature dependences:
  - k_{V} depends on the true activation energy E_{act} of the reaction.
  - D_{e} has only a very weak temperature dependence (except for the special case of strong adsorption sites when surface diffusion becomes a limiting factor)
- We can see that the apparent activation energy will be $E_{\text{act}}/2$ (unless surface diffusion is a limiting factor).

6.2.8 Allowing for external mass transport limitations

- At steady state, we expect

  Rate of transport of reagent to surface = rate of reaction in catalyst



- For a first-order reaction we can write (using external mass transport coefficient k_{g})

$$k_{\text{g}}(C_{\text{B}} - C_{\text{S}})A_{\text{ext}} = \eta k_{\text{V}}C_{\text{S}}V_{\text{cat}}$$

$$C_{\text{S}} = \frac{C_{\text{B}}}{1 + \dfrac{\eta k_{\text{V}}V_{\text{cat}}}{k_{\text{g}}A_{\text{ext}}}}$$

- The overall rate in mol/s is then

$$r_{\text{overall}} = \eta k_{\text{V}} V_{\text{cat}} \frac{C_{\text{B}}}{1 + \frac{\eta k_{\text{V}} V_{\text{cat}}}{k_{\text{g}} A_{\text{ext}}}}$$

- Sometimes we define an **overall** effectiveness factor.

$$\Omega = \frac{\text{average rate of reaction with (heat and) mass transfer limitations}}{\text{rate of reaction evaluated at } \textbf{bulk} \text{ conditions}}$$

$$\Omega = \frac{\text{mol/s reacting in catalyst}}{\text{mol/s that would react if whole catalyst was at } \textbf{bulk} \text{ conditions } (T_{\text{B}}, C_{\text{B}})}$$

$$\Omega = \frac{r_{\text{overall}}}{r_{\text{no transport limitations}}}$$

- For a first-order reaction this gives

$$\Omega = \frac{\eta}{1 + \frac{\eta k_{\text{V}} V_{\text{cat}}}{k_{\text{g}} A_{\text{ext}}}}$$

- Note that if there is minimal internal diffusion resistance (so $\eta = 1$), the overall effectiveness factor is

$$\Omega = \frac{1}{1 + \frac{k_{\text{V}} V_{\text{cat}}}{k_{\text{g}} A_{\text{ext}}}}$$

This is unusual.

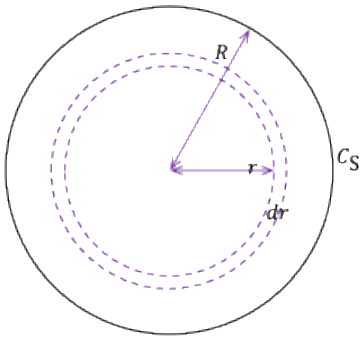
- The dimensionless parameter $k_{\text{V}} V_{\text{cat}} / k_{\text{g}} A_{\text{ext}}$ quantifies the relative rates of chemical reaction and external mass transport.

6.3 Thermal effects in porous catalysts

- Up until now we have assumed that catalytic reactions are isothermal.
- The isothermal approximation for catalyst particles will break down when the reaction rate becomes so large that the heat generated (or consumed) by reaction cannot be dissipated (or supplied) at a rate that is sufficient to keep the entire catalyst particle at the same temperature as the surrounding fluid.
- Temperature gradients can exist within the particle itself or within the stagnant film separating the particle and the surrounding fluid. We can analyse the effects of temperature gradients in terms of catalyst effectiveness factors.
- The variation of concentration and temperature throughout the catalyst will be coupled.

6.3.1 Intra-Particle Temperature Gradients

For a spherical catalyst particle and an nth order reaction, the mass balance at steady state (following the approach of section 6.2.3) is:



$$-\frac{d}{dr}\left(-4\pi r^2 D_e \frac{dC}{dr}\right)dr - k_V C^n 4\pi r^2 dr = 0$$

$$D_e\left(\frac{d^2C}{dr^2} + \frac{2}{r}\frac{dC}{dr}\right) = k_V C^n$$

Analogously, the heat balance at steady state is:

$$-\frac{d}{dr}\left(-4\pi r^2 \lambda_{eff} \frac{dT}{dr}\right)dr - \Delta H_r k_V C^n 4\pi r^2 dr = 0$$

$$\lambda_{eff}\left(\frac{d^2T}{dr^2} + \frac{2}{r}\frac{dT}{dr}\right) = \Delta H_r k_V C^n$$

λ_{eff} = effective thermal conductivity

$\lambda_{eff} \approx \lambda_s \exp\left[-1.5\varepsilon_{particle}/\left(1-\varepsilon_{particle}\right)\right]$

[J. Eur. Ceram. Soc. 29 (2009) 347-353]

λ_s = thermal conductivity of solid (support)

T = temperature at radius r

It is convenient to define the rate constant at any position (and so temperature) k_V relative to the rate constant at the surface of the particle k_{VS}:

$$\frac{k_V}{k_{VS}} = \frac{A\exp\left(\frac{-E_a}{RT}\right)}{A\exp\left(\frac{-E_a}{RT_S}\right)} = \exp\left(\frac{-E_a}{R}\left[\frac{1}{T} - \frac{1}{T_S}\right]\right)$$

A = pre-exponential factor

E_a = activation energy of reaction

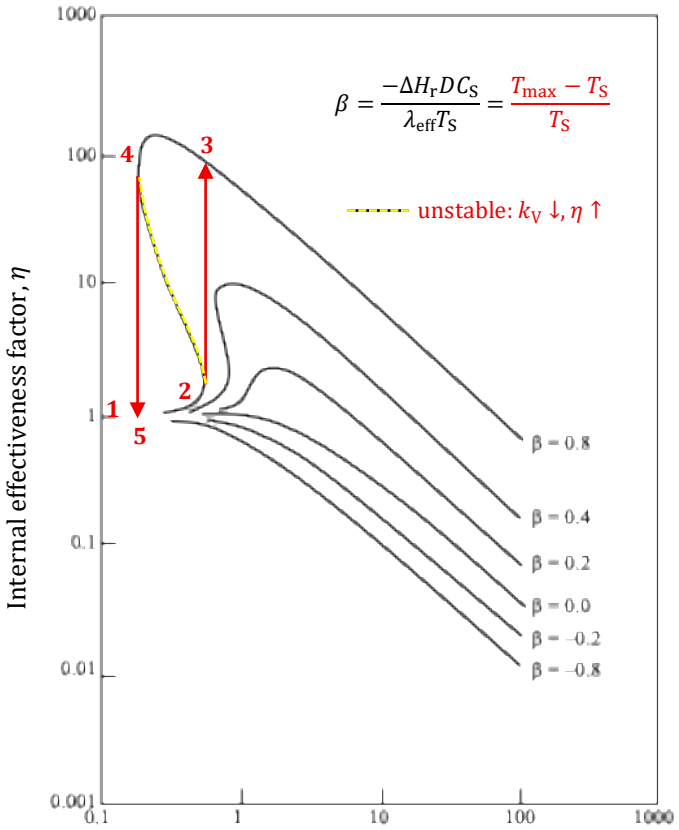
T_S = temperature at surface of particle

The internal effectiveness factor is then:

$$\eta = \frac{k_{VS}\int_0^R 4\pi r^2 C^n \exp\left(\frac{-E_a}{R}\left[\frac{1}{T}-\frac{1}{T_S}\right]\right)dr}{k_{VS}4\pi R^3 C_S^n} = \int_0^R \left(\frac{r}{R}\right)^2 \left(\frac{C}{C_S}\right)^n \exp\left(\frac{-E_a}{R}\left[\frac{1}{T}-\frac{1}{T_S}\right]\right)\frac{dr}{R}$$

In order to determine η, we need to know the concentration and temperature at all points in the catalyst pellet. To obtain these, we need to simultaneously solve the two coupled differential equations above. This is not straightforward because D_e, ΔH_r and λ_{eff} are all functions of temperature (and therefore position), as well as k_V. D_e, ΔH_r and λ_{eff} are fairly weak functions of temperature and so it is often assumed that they are independent of temperature, which somewhat simplifies the problem, but a numerical solution is nonetheless required.

Solutions for a first order reaction in a spherical particle, with $\frac{E_a}{RT_S} = 20$ are shown below [adapted from Weisz & Hicks, *Chem Eng Sci* 17 (1962) 265-275]:



- $\eta > 1$ is possible
- $\beta = 0.8$ is very large
- For large β, multiple steady states may occur

Hysteresis may occur where multiple steady states are possible:

1. Start reaction (exothermic)
2. Reaction gives out heat $\Longrightarrow T$ increases $\Longrightarrow k_V$ increases $\Longrightarrow \varphi_{1,gen} = \frac{V_{cat}}{A_{ext}}\sqrt{\frac{k_V}{D_e}}$ increases

(probably external heat transfer limited now)

3. Jump to (much) higher η
4. Catalyst now operates at much higher rate $\Longrightarrow$ uses up reactants (probably external mass transfer limited now) $\Longrightarrow$ rate decreases $\Longrightarrow T$ decreases again
5. Jump back to $\eta \approx 1$

This will result in large (>100°C), rapid temperature jump and unstable operation, both highly undesirable: therefore want to avoid operating in this regime.

Prater's analysis

Although it is difficult to obtain the temperature and concentration profiles in non-isothermal cases, it is possible to simply obtain a criterion for whether the intra-particle temperature gradient is significant (i.e is β significantly > 0), following the analysis of Prater (1958).

For nth order reaction in a spherical porous catalyst particle, we have already seen that:

$$D_e\left(\frac{d}{dr}\left\{r^2\frac{dC}{dr}\right\}\right) = r^2 k_V C^n$$

$$\lambda_{eff}\left(\frac{d}{dr}\left\{r^2\frac{dT}{dr}\right\}\right) = r^2 \Delta H_r k_V C^n$$

Eliminating $r^2 k_V C^n$:

$$D_e\left(\frac{d}{dr}\left\{r^2\frac{dC}{dr}\right\}\right) = \frac{\lambda_{eff}}{\Delta H_r}\left(\frac{d}{dr}\left\{r^2\frac{dT}{dr}\right\}\right)$$

$$\left(\frac{d}{dr}\left\{r^2\frac{d\alpha C}{dr}\right\}\right) = \left(\frac{d}{dr}\left\{r^2\frac{dT}{dr}\right\}\right) \qquad \alpha = \frac{\Delta H_r D_e}{\lambda_{eff}}$$

$$\left(\frac{d}{dr}\left\{r^2\frac{d[\alpha C - T]}{dr}\right\}\right) = 0$$

Integrating:

$$r^2\frac{d[\alpha C - T]}{dr} = a$$

$$\frac{d[\alpha C - T]}{dr} = \frac{a}{r^2}$$

Integrating again:

$$\alpha C - T = -\frac{a}{r} + b$$

Boundary conditions:

At $r = 0$: $\quad \frac{dC}{dr} = \frac{dT}{dr} = 0 \quad \Longrightarrow a = 0$

At $r = R$: $\quad C = C_S$ and $T = T_S \quad \Longrightarrow b = \alpha C_S - T_S$

So:

$$T - T_S = \alpha(C - C_S) = \frac{-\Delta H_r D_e}{\lambda_{eff}}(C_S - C)$$

[We have shown this to be the case only a spherical particle, but it is straightforward to show it is true for any particle shape]

The maximum possible value of T occurs when $C = 0$.

Thus the maximum value of β is given by:

$$\beta = \frac{T_{max} - T_S}{T_S} = \frac{-\Delta H_r D_e C_S}{\lambda_{eff} T_S}$$

(for an exothermic reaction: the maximum possible value of C_S is C_B, the reactant concentration in the bulk fluid far from the catalyst particle; and the minimum possible value of T_S is T_B, the temperature of the bulk fluid far from the catalyst particle)

Therefore:

$$\beta \leq \frac{-\Delta H_r D_e C_B}{\lambda_{eff} T_B}$$

Internal temperature variations may reasonably be neglected if $\beta \lesssim 0.02$ say

So without needing to solve for C and T as a function of position (which is difficult), we have a criterion for determining whether the internal variations in temperature are significant. If they are not, we can use the isothermal analysis of section 6.2. Note however there may still be an **external** temperature gradient (section 6.3.2) and so, although uniform, the particle temperature may be lower or higher than that of the bulk fluid. This is in fact quite common, particularly for gases, since the thermal conductivity of gases is much lower than that of solids.

6.3.2 External Temperature Gradients

At steady state, the heat flow from the bulk fluid into (endothermic reaction) or out of (exothermic reaction) the particle must be equal to the heat consumed or generated by reaction within the catalyst particle.

Thus for a particle of arbitrary shape, and nth order reaction:

$$A_{ext} h(T_B - T_S) = V_{cat} \eta k_V C_S^n (-\Delta H_r)$$

where T_B and T_S and the temperature in the bulk fluid and at the particle surface, respectively; and h is the heat transfer coefficient from the fluid to the catalyst particle.

$$\Delta T_{ext} = (T_B - T_S) = \left(\frac{V_{cat}}{A_{ext}}\right)\frac{\eta k_V C_S^n (-\Delta H_r)}{h}$$

$\left(\frac{V_{cat}}{A_{ext}}\right)$ is the same length dimension we used for the generalised Thiele modulus (L for a slab of thickness $2L$; $R/3$ for a sphere or radius R).

To determine whether the external temperature gradient is insignificant, we may assume $\eta = 1$, $C_S = C_B$ and obtain h from a suitable correlation, for example the Wakao and Kaguei correlation [Wakao, N., & Kaguei, S. (1982) *Heat and Mass Transfer in Packed Beds*, Gordon & Breach Science Publishers, New York]:

$$Nu = 2 + 1.1Re^{0.6}Pr^{1/3} \qquad \text{for packed beds with } 10 \leq Re \leq 10^4$$

Here Nu and Re are based on the particle dimension (diameter for spheres).

If $\frac{\Delta T_{ext}}{T_S} \lesssim 0.02$, say, then external heat transfer may be safely ignored.

Often the temperature gradient within the particle is small and external concentration gradient is also small, so η may be obtained using the isothermal methods of Section 6.2, and $C_S \approx C_B$. Although isothermal, the catalyst particle may be hotter (exothermic reaction) or cooler (endothermic reaction) than the bulk fluid. In general , we therefore need an iterative solution to determine T_S and η because k_V is a strong function of T_S via the Arrhenius equation.
The overall effectiveness factor is:

$$\Omega = \frac{k_{VS}}{k_{VB}}\eta = \eta \exp\left(\frac{E_a}{R}\left[\frac{1}{T_B} - \frac{1}{T_S}\right]\right)$$

where k_{VS} and k_{VB} are the reaction rate constants at T_S and T_B, respectively.

In the low Thiele number regime, $\eta \approx 1$, and so it is straightforward to evaluate Ω.

In the high Thiele number regime,

$$\eta \approx \frac{1}{\varphi_{n,gen}} \qquad \varphi_{n,gen} = \frac{V_{cat}}{A_{ext}}\sqrt{\frac{(n+1)k_{VS}C_S^{n-1}}{2D_e}}$$

Note that k_{VS} must be evaluated as surface conditions (i.e. at T_S), making Ω harder to evaluate in this regime.

An isothermal catalyst particle behaves a bit like a miniature CSTR. In the same way as for a CSTR, multiple steady states are possible for exothermic reactions – one at low temperature, where the rate of reaction and heat transfer are both low; one at high temperature, where reaction rate and heat transfer are both high; and an unstable one in between. In the case of a CSTR the upper bound on reaction rate is set by 100% conversion (i.e. reaction rate cannot increase indefinitely with temperature). In the case of a catalyst pellet, reaction rate is ultimately limited by external mass transfer when the particle gets very hot:

$$k_g(C_B - C_S)\Delta H_r = h(T_B - T_S)$$

When T_S becomes very large, $C_S \to 0$.
For gases $k_g \approx h$ (see Section 2.2.5).

Thus the maximum T_S (and so rate constant) is set by:

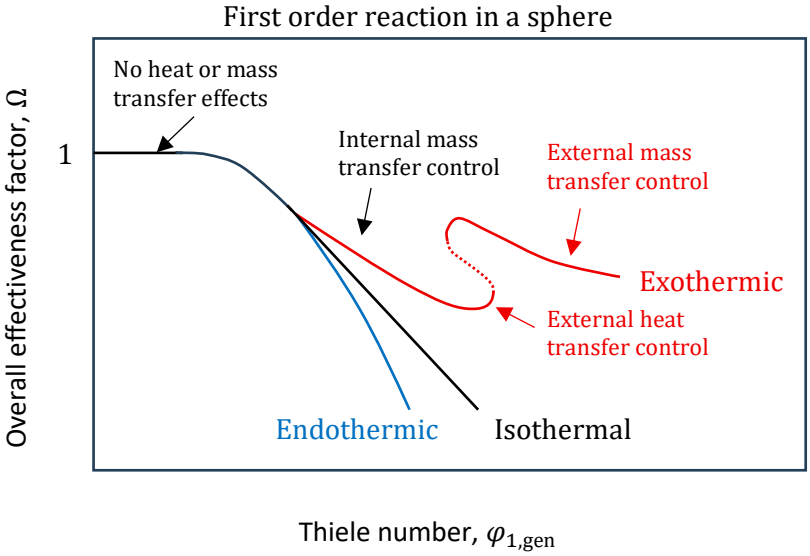
$$T_S = T_B - C_B \Delta H_r$$

In this external mass transfer regime, for a first order reaction in a spherical particle:

$$\Omega = \frac{4\pi R^2 k_g (C_B - C_S)}{\frac{4}{3}\pi R^3 k_{VB} C_B} \approx \frac{3}{R}\frac{k_g}{k_{VB}}$$

c.f. $\Omega \approx \eta \approx \frac{3}{R}\sqrt{\frac{k_g}{k_{VB}}}$ in the internal diffusion limited regime

The sketch below illustrates the type of behaviour which is possible when external heat transfer resistance is significant, but internal heat transfer resistance is not.



- $\eta > 1$ is possible
- Multiple steady states may occur
- Unstable operation and large T jumps can then occur
- This is undesirable